

MINERAL AND ENERGY RESOURCES: WORLD AND INDIA

GEOLOGICAL CONTROL OF RESOURCE → MINERAL AND ENERGY CLASSIFICATION → WORLD MINERAL-ENERGY BELT RAIL → EXTRACTION TO INDUSTRY CHAIN → OCCURRENCE TO USABLE RESOURCE → FOUR-ELEMENT PETROLEUM SYSTEM

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 GEOLOGICAL CONTROL OF RESOURCE DISTRIBUTION

GEOLOGICAL CONTROL OF RESOURCE DISTRIBUTION

GEOLOGICAL HISTORY

ROCK STRUCTURE, BASINS, TECTONICS, SURFACE WEATHERING

IGNEOUS-METAMORPHIC TERRAIN

METALLIC ORES

SEDIMENTARY BASIN

COAL, PETROLEUM, NATURAL GAS

TROPICAL LATERITISATION

GEOLOGICAL HISTORY → rock structure, basins, tectonics, surface weathering

SHIELD / IGNEOUS-METAMORPHIC TERRAIN → metallic ores

SEDIMENTARY BASIN → coal, petroleum, natural gas

TROPICAL LATERITISATION → bauxite on suitable plateaux and hilltops

TRAP → distribution is a deduction from Earth history, not a country list

DEFINITION / COMPONENTS

- GEOLOGICAL HISTORY → rock structure, basins, tectonics, surface weathering
- SHIELD / IGNEOUS-METAMORPHIC TERRAIN → metallic ores

TRIGGER / MECHANISM

- SEDIMENTARY BASIN → coal, petroleum, natural gas
- TROPICAL LATERITISATION → bauxite on suitable plateaux and hilltops

EFFECT / INTERVENTION

- TRAP → distribution is a deduction from Earth history, not a country list
- GEOLOGICAL HISTORY → rock structure, basins, tectonics, surface weathering

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → distribution is a deduction from Earth history, not a country list.

01

STAGE 01 MINERAL AND ENERGY CLASSIFICATION LADDER

MINERAL AND ENERGY CLASSIFICATION LADDER

FERROUS METALLIC

IRON ORE, MANGANESE, CHROMITE; STEEL-MAKING BASE

NON-FERROUS METALLIC

COPPER, BAUXITE, LEAD, ZINC

NON-METALLIC INDUSTRIAL

MICA, LIMESTONE, GYPSUM, PHOSPHATES

MINERAL FUELS

CLASSIFICATION

FERROUS METALLIC → iron ore, manganese, chromite; steel-making base
NON-FERROUS METALLIC → copper, bauxite, lead, zinc

DIAGNOSTIC BASIS

NON-METALLIC INDUSTRIAL → mica, limestone, gypsum, phosphates
MINERAL FUELS → coal, petroleum, natural gas, uranium

PROCESS LINK

ENERGY PAIRS → conventional vs non-conventional; exhaustible stock vs renewable flow
FERROUS METALLIC → iron ore, manganese, chromite; steel-making base

EXAM TRAP

NON-FERROUS METALLIC → copper, bauxite, lead, zinc
NON-METALLIC INDUSTRIAL → mica, limestone, gypsum, phosphates

CLASSIFICATION

- FERROUS METALLIC → iron ore, manganese, chromite; steel-making base
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EXAM TRAP

- NON-FERROUS METALLIC → copper, bauxite, lead, zinc
- NON-METALLIC INDUSTRIAL → mica, limestone, gypsum, phosphates

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENERGY PAIRS → conventional vs non-conventional; exhaustible stock vs renewable flow.

02

STAGE 02 WORLD MINERAL-ENERGY BELT RAIL

WORLD MINERAL-ENERGY BELT RAIL

PERSIAN GULF

CLASSIC SEDIMENTARY HYDROCARBON PROVINCE

INTERIOR USA

COAL, OIL, GAS WITH IRON ASSOCIATIONS

HISTORIC COAL-IRON HEAVY-INDUSTRY CORE

CHINESE INTERIOR

CONTINENTAL-SCALE FUEL FRONTIERS

PERSIAN GULF → classic sedimentary hydrocarbon province

APPALACHIAN / INTERIOR USA → coal, oil, gas with iron associations

RUHR-LORRAINE → historic coal-iron heavy-industry core

RUSSIAN-SIBERIAN / CHINESE INTERIOR → continental-scale fuel frontiers

DEFINITION / COMPONENTS

- PERSIAN GULF → classic sedimentary hydrocarbon province
- APPALACHIAN / INTERIOR USA → coal, oil, gas with iron associations

TRIGGER / MECHANISM

- RUHR-LORRAINE → historic coal-iron heavy-industry core
- RUSSIAN-SIBERIAN / CHINESE INTERIOR → continental-scale fuel frontiers

EFFECT / INTERVENTION

- COPPERBELT, AUSTRALIAN SHIELD, ANDES → shield and tectonic metal provinces
- PERSIAN GULF → classic sedimentary hydrocarbon province

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: COPPERBELT, AUSTRALIAN SHIELD, ANDES → shield and tectonic metal provinces.

03

STAGE 03 EXTRACTION TO INDUSTRY CHAIN

EXTRACTION TO INDUSTRY CHAIN

ORE OR FUEL RAISED AT THE DEPOSIT

RAIL, SLURRY PIPELINE, PORT OR INLAND WATERWAY EVACUATION

BENEFICIATION, SMELTING, REFINING, PETROCHEMICAL CONVERSION

POWER + MARKET

DISPATCHABLE ENERGY, CAPITAL AND DEMAND COMPLETE THE CLUSTER

A PRODUCING REGION IS NOT AUTOMATICALLY AN INDUSTRIAL REGION

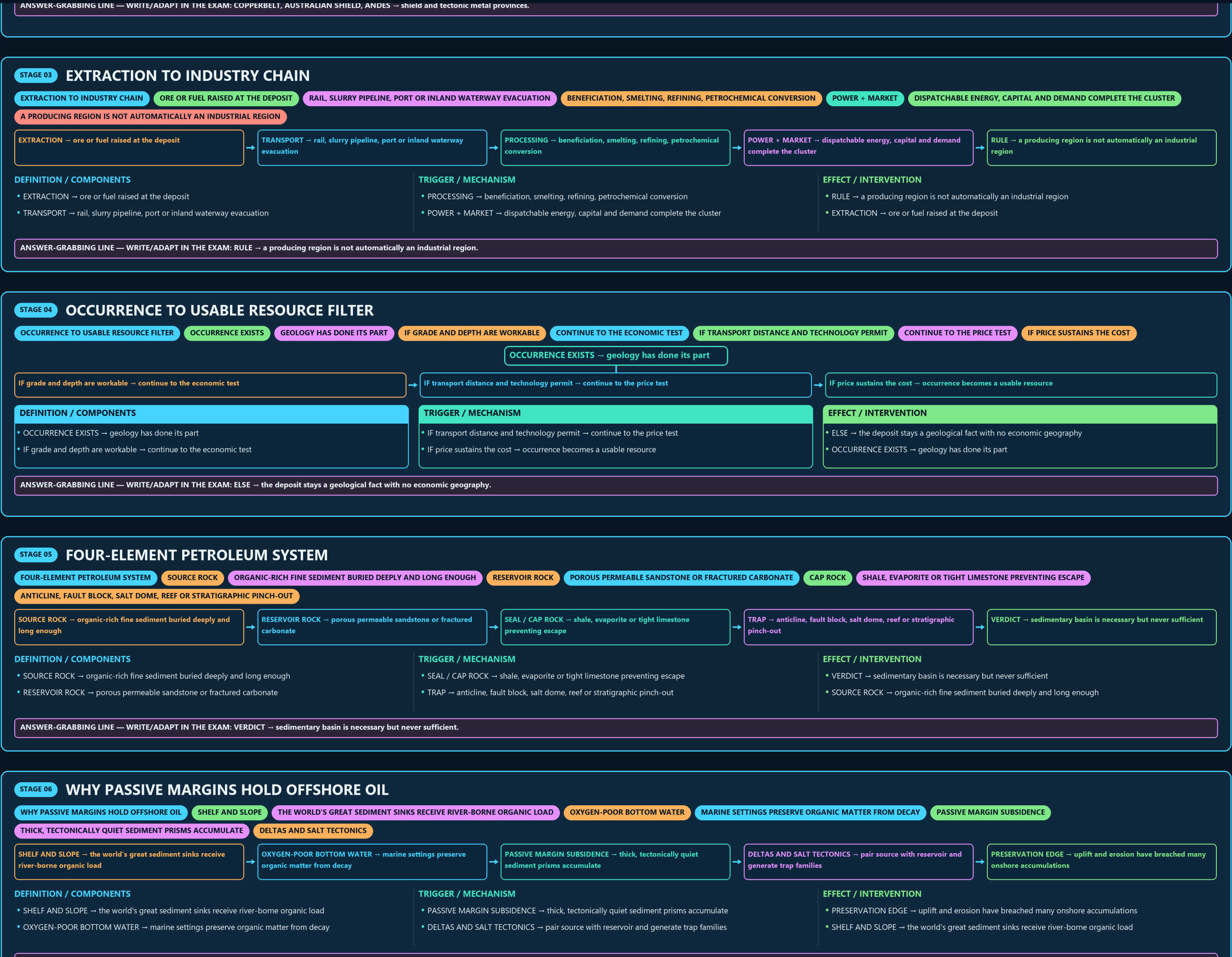
EXTRACTION → ore or fuel raised at the deposit

TRANSPORT → rail, slurry pipeline, port or inland waterway evacuation

PROCESSING → beneficiation, smelting, refining, petrochemical conversion

POWER + MARKET → dispatchable energy, capital and demand complete the cluster

RULE → a producing region is not automatically an industrial region



WHY PASSIVE MARGINS HOLD OFFSHORE OIL

SHELF AND SLOPE

THE WORLD'S GREAT SEDIMENT SINKS RECEIVE RIVER-BORNE ORGANIC LOAD

OXYGEN-POOR BOTTOM WATER

MARINE SETTINGS PRESERVE ORGANIC MATTER FROM DECAY

PASSIVE MARGIN SUBSIDENCE

THICK, TECTONICALLY QUIET SEDIMENT PRISMS ACCUMULATE

DELTA AND SALT TECTONICS

SHELF AND SLOPE → the world's great sediment sinks receive river-borne organic load

OXYGEN-POOR BOTTOM WATER → marine settings preserve organic matter from decay

PASSIVE MARGIN SUBSIDENCE → thick, tectonically quiet sediment prisms accumulate

DELTA AND SALT TECTONICS → pair source with reservoir and generate trap families

PRESERVATION EDGE → uplift and erosion have breached many onshore accumulations

DEFINITION / COMPONENTS

- SHELF AND SLOPE → the world's great sediment sinks receive river-borne organic load
- OXYGEN-POOR BOTTOM WATER → marine settings preserve organic matter from decay

TRIGGER / MECHANISM

- PASSIVE MARGIN SUBSIDENCE → thick, tectonically quiet sediment prisms accumulate
- DELTA AND SALT TECTONICS → pair source with reservoir and generate trap families

EFFECT / INTERVENTION

- PRESERVATION EDGE → uplift and erosion have breached many onshore accumulations
- SHELF AND SLOPE → the world's great sediment sinks receive river-borne organic load

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DELTA AND SALT TECTONICS → pair source with reservoir and generate trap families.

07

STAGE 07

OFFSHORE VERSUS ONSHORE COMPARISON

OFFSHORE VERSUS ONSHORE COMPARISON

ONSHORE INTRACRATONIC AND RIFT BASINS VS SHELF, SLOPE AND

COST AND TECHNOLOGY

INCREMENTAL LAND DRILLING VS LUMPY HIGH-COST MARINE SYSTEMS

SMALL ONSHORE FIELDS CAN BE VIABLE; OFFSHORE NEEDS LARGE

LAND ACQUISITION ONSHORE VS FISHERIES, SHIPPING AND SPILL RISK

ORDINARY NATIONAL LAW VS MARITIME ZONES AND CONTESTED BOUNDARIES

DEFINITION / COMPONENTS

TRIGGER / MECHANISM

EFFECT / INTERVENTION

GEOLOGY → onshore intracratonic and rift basins vs shelf, slope and deepwater margin

THRESHOLD → small onshore fields can be viable; offshore needs large fields

JURISDICTION → ordinary national law vs maritime zones and contested boundaries

COST AND TECHNOLOGY → incremental land drilling vs lumpy high-cost marine systems

FOOTPRINT → land acquisition onshore vs fisheries, shipping and spill risk offshore

GEOLOGY → onshore intracratonic and rift basins vs shelf, slope and deepwater margin

DEFINITION / COMPONENTS

- GEOLOGY → onshore intracratonic and rift basins vs shelf, slope and deepwater margin
- COST AND TECHNOLOGY → incremental land drilling vs lumpy high-cost marine systems

TRIGGER / MECHANISM

- THRESHOLD → small onshore fields can be viable; offshore needs large fields
- FOOTPRINT → land acquisition onshore vs fisheries, shipping and spill risk offshore

EFFECT / INTERVENTION

- JURISDICTION → ordinary national law vs maritime zones and contested boundaries
- GEOLOGY → onshore intracratonic and rift basins vs shelf, slope and deepwater margin

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: JURISDICTION → ordinary national law vs maritime zones and contested boundaries.

08

STAGE 08

INDIA PETROLEUM PROVINCE MAP HOOKS

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ASSAM-ARAKAN BASIN AND THE CAMBAY BASIN OF WESTERN INDIA

WESTERN OFFSHORE

MAJOR CONTINENTAL-MARGIN PROVINCE, CARBONATE RESERVOIR CHARACTER

EAST-COAST OFFSHORE

KRISHNA-GODAVARI AND CAUVERY DELTAIC MARGINS, GAS-PRONE

COASTAL CONSEQUENCE

LANDFALL TERMINALS PULL REFINING AND PETROCHEMICALS TO THE COAST

SYSTEM / DEFINITION

PROCESS / MECHANISM

SPATIAL / INDIA PATTERN

HAZARD / LIMIT / RESPONSE

ONSHORE → Assam-Arakan basin and the Cambay basin of western India

EAST-COAST OFFSHORE → Krishna-Godavari and Cauvery deltaic margins, gas-prone

CAUTION → name provinces and reservoir character, never remembered output figures

WESTERN OFFSHORE → major continental-margin province, carbonate reservoir character

COASTAL CONSEQUENCE → landfall terminals pull refining and petrochemicals to the coast

ONSHORE → Assam-Arakan basin and the Cambay basin of western India

EAST-COAST OFFSHORE → Krishna-Godavari and Cauvery deltaic margins, gas-prone

SYSTEM / DEFINITION

- ONSHORE → Assam-Arakan basin and the Cambay basin of western India
- WESTERN OFFSHORE → major continental-margin province, carbonate reservoir character

PROCESS / MECHANISM

- EAST-COAST OFFSHORE → Krishna-Godavari and Cauvery deltaic margins, gas-prone
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SPATIAL / INDIA PATTERN

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09

STAGE 09

CRITICAL-MINERAL CHOKEPOINT FORK

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TRANSITION DEMAND RISES

LITHIUM, COBALT, NICKEL, GRAPHITE, COPPER, RARE EARTHS

IF THE QUESTION IS EXTRACTION

SUPPLY IS GEOLOGICALLY NARROW BUT VISIBLE

IF THE QUESTION IS PROCESSING

CONCENTRATION IS TIGHTER AND OFTEN IN OTHER COUNTRIES

THE BINDING CHOKEPOINT IS MIDSTREAM, NOT AT THE MINE

TRANSITION DEMAND RISES → lithium, cobalt, nickel, graphite, copper, rare earths

IF the question is extraction → supply is geologically narrow but visible

IF the question is processing → concentration is tighter and often in other countries

THEREFORE → the binding chokepoint is midstream, not at the mine

DEFINITION / COMPONENTS

- TRANSITION DEMAND RISES → lithium, cobalt, nickel, graphite, copper, rare earths
- IF the question is extraction → supply is geologically narrow but visible

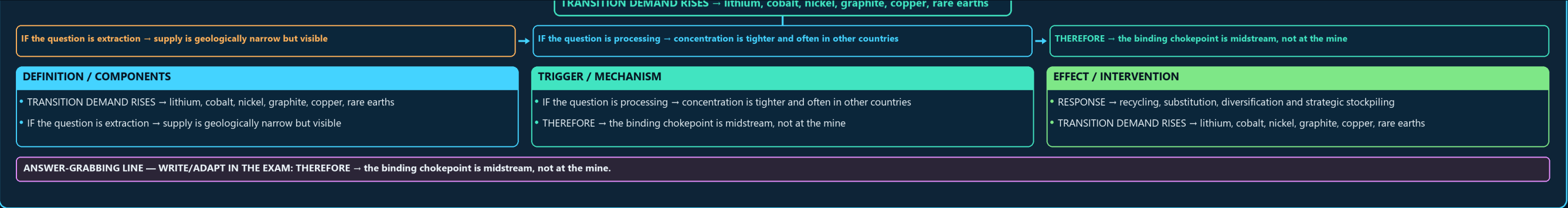
TRIGGER / MECHANISM

- IF the question is processing → concentration is tighter and often in other countries
- THEREFORE → the binding chokepoint is midstream, not at the mine

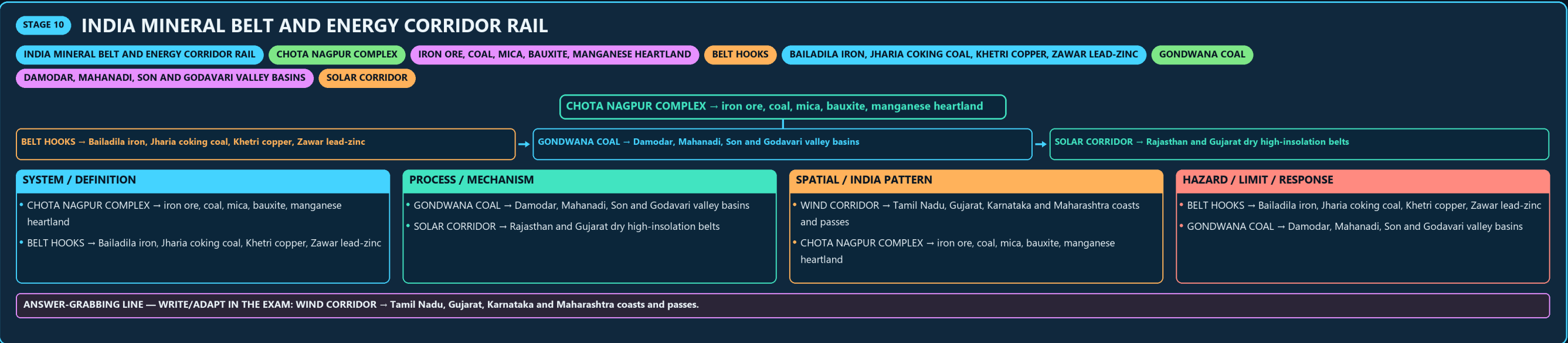
EFFECT / INTERVENTION

- RESPONSE → recycling, substitution, diversification and strategic stockpiling
- TRANSITION DEMAND RISES → lithium, cobalt, nickel, graphite, copper, rare earths

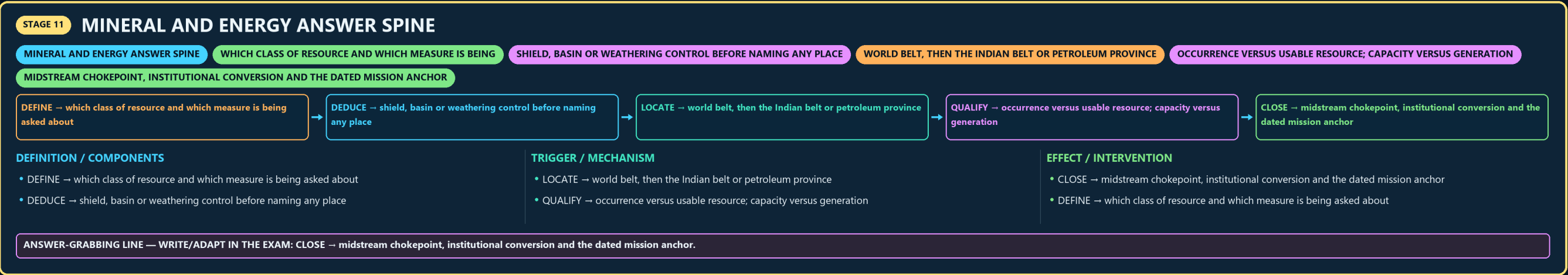
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: THEREFORE → the binding chokepoint is midstream, not at the mine.



10



11



E

